British Applied College
الكلية التطبيقية البريطانية

Threats to Data, Information, & Systems

Comprehensive Overview and Impacts

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Learning Objectives

01 **Define and categorize** the characteristics of various digital threats, including malware and social engineering.

02 **Analyze** the multifaceted impact of cyberattacks on individuals, focusing on financial and psychological consequences.

03 **Evaluate** the operational, financial, and legal risks organizations face when data security is compromised.

04 **Engage** in interactive discussions and scenario-based activities to apply theoretical cybersecurity knowledge.

Class Engagement

Interactive Icebreaker

"What is the most common cyber threat you have encountered or heard about in the news lately?"

Characteristics of Malware

Core Definition

Malware is a type of software specifically designed to harm, exploit, or gain unauthorized access to a computer system.

Key Classifications

Includes viruses, worms, Trojans, and other malicious software variants.

Infection Vectors

Exploiting system vulnerabilities or tricking users into executing malicious code.

Primary Impacts

Once infected, systems suffer from data loss, theft, or widespread corruption.

System Control

Malware can grant attackers remote control or silent access to sensitive data.

The Human Element: Hackers

Unauthorized access to computer systems or networks by individuals or groups.

Common Techniques

Brute force attacks (password guessing)

Social engineering (psychological manipulation)

Exploiting software vulnerabilities

Post-Access Actions

Stealing sensitive or personal data

Modifying or deleting critical files

Causing widespread system damage

Phishing:
Social Engineering

The Art of Deception

Core Concept

Deceptive tactics used to trick users into revealing sensitive data through psychological manipulation.

Target Data

Usernames, passwords, credit card details, and other personally identifiable information (PII).

Communication Channels

Carried out via email, text messages (smishing), or phone calls (vishing).

Source Appearance

Messages often mimic legitimate and trusted sources like banks, IT departments, or government agencies.

Practical Application

Spot the Phish

Challenge: Identify the Red Flags

Examine the visual cues on the right. What makes this communication suspicious?

- Suspicious Sender Address
- Generic Greetings
- Urgent or Threatening Tone
- Mismatched or Spoofed URLs

Accidental Damage

The Unseen Threat

Common Causes

- Hardware failure (Disk crashes, component wear)
- Power outages and surges
- Human error (Accidental deletion, liquid spills)

Direct Impact

- Permanent data loss
- File corruption and integrity issues
- System inaccessibility and downtime

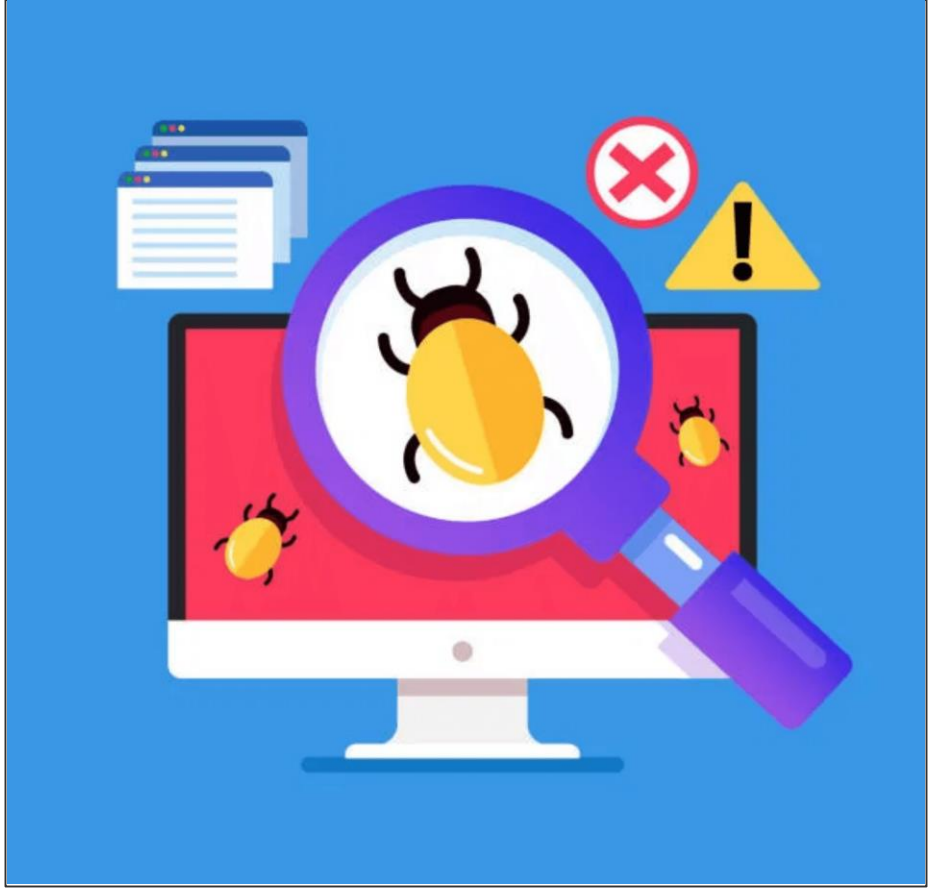
Mitigation

- Regular backup and recovery procedures
- Uninterruptible Power Supplies (UPS)
- Comprehensive employee training

Collaborative Learning

Think-Pair-Share

Prompt: "How can you tell if your device is infected with malware?"



Impact on Individuals

Financial & Identity Consequences

Financial Loss

- Unauthorized access to bank accounts or credit cards.
- Theft of funds or fraudulent purchases.
- Significant damage to personal credit scores.

Identity Theft

- Cybercriminals stealing personal identity for fraud.
- Opening new credit accounts or taking out loans.
- Extremely damaging and time-consuming to resolve.

Privacy, Reputation, and Emotion

The Psychological and Social Impact

Privacy Violations

- Compromise of Social Security Numbers (SSNs)
- Exposure of sensitive medical records
- Leaking of private conversations and chats

Reputation Damage

- Leaked photos or private info harming relationships
- Professional credibility loss due to data leaks
- Long-lasting effects on personal and social image

Emotional Distress

- Feelings of fear, anger, or helplessness
- Traumatic experience of being a cyberattack victim
- Loss of access to critical work or personal files

Impact on Organisations

Financial & Reputation

Financial Loss

- Direct losses: Lost revenue, equipment damage, and legal fees.
- Intellectual Property (IP): Theft of trade secrets and patents.
- Long-term costs: Decreased revenue and market share.

Reputational Damage

- Brand Image: Negative publicity and loss of customer trust.
- Customer Churn: Customers moving to competitors.
- Trust Recovery: Difficult and time-consuming to regain.

Scenario-Based Analysis

Case Study Mini-Activity

The Scenario

"A major international retail chain suffers a massive data breach, resulting in the loss of over 1 million customer credit card records and personal details."

Your Task

In your groups, list the **top 3 immediate consequences** for the organization.

Consider financial, reputational, and legal perspectives.

Time: 10 Minutes

Legal, Regulatory, and Operational Impacts

Compliance Risks and Business Continuity

Legal & Regulatory

- Substantial fines and regulatory penalties.
- Lawsuits from affected customers or partners.
- Non-compliance with data protection laws (e.g., GDPR).

Operational Disruption

- System downtime leading to lost productivity.
- Delays in service delivery and missed deadlines.
- Damage to critical infrastructure and equipment.

Market Impact

- Competitive disadvantage from IP theft.
- Decreased revenue and loss of market share.
- Long-term erosion of customer trust and loyalty.

Summary & Conclusion

01 **Threat Diversity:** Cybersecurity threats are multifaceted, ranging from technical malware to psychological social engineering and accidental human error.

02 **Individual Impact:** Victims face severe financial loss, identity theft, privacy violations, and significant emotional distress.

03 **Organisational Risk:** Businesses risk financial ruin, intellectual property theft, legal penalties, and the long-term loss of customer trust.

"Security is a shared responsibility."

Final Knowledge Check

Reviewing Key Concepts

01 What is the primary technical difference between a virus and a phishing attack?

02 Identify one critical strategy to mitigate the impact of accidental data damage.

03 Why is reputational damage considered one of the most difficult consequences for an organization to recover from?